SEC 1 · SATELLITE TYPES & ORBITS

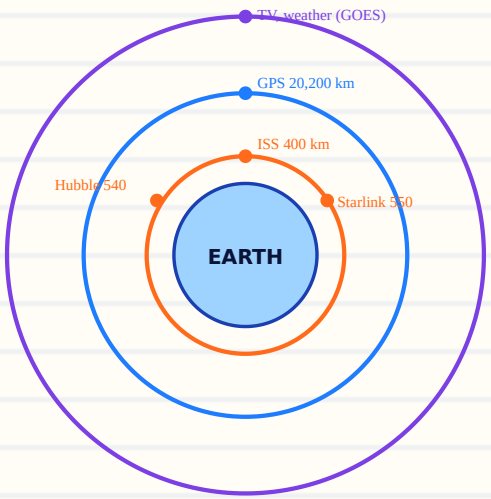
Your spread, upgraded

Your right page was already the correct shape. This is the same content with a drawn left page, harder cues, grouped lists, and the summary box. Copy what helps.

LEFT PAGE · DRAW

SEC 1 — SATELLITE TYPES & ORBITS

date · pg __



GEO = 35,786 km
1 orbit = 1 day → hangs over one spot
(geoSTATIONARY if also over the equator)

MEO = 2,000 – 35,786 km
GPS: 12-hour orbit

LEO = 180 – 2,000 km
~90-minute orbit, 7.7 km/s
drag + atomic oxygen live here

rule: higher = slower = longer day
LEO 90 min → MEO 12 h → GEO 24 h

not to scale! GEO is ~6 Earth-radii out →

⭐ 3 TAKEAWAYS

1. Higher orbit = slower = longer trip around (90 min → 12 h → 24 h)
2. GEO is special only because its day matches Earth's day
3. Group the 8 satellite types by JOB: look out / look down / relay / live

RIGHT PAGE · CORNELL

CUES = questions | **NOTES = keywords**

Which satellite types **look out**, which **look down**, which **relay**?

Where does the ISS live? GPS? TV satellites?

Why is GEO exactly 35,786 km?

Geosynchronous vs geostationary?

Higher orbit → faster or slower?

? what orbit do spy sats use, and why

LOOK OUT — astronomical (Hubble, JWST)
LOOK DOWN — Earth obs (Landsat), weather (GOES), recon (spy)
RELAY — comms (Starlink, DirecTV), navigation (GPS)
LIVE — biosatellites (Laika!), space stations (ISS)

LEO 180–2000 km · ISS 400 · ~90 min
MEO 2000–35,786 · GPS 20,200 · 12 h
GEO 35,786 · 1 orbit = 1 Earth day → stays put
geoSYNC = 1-day orbit, any tilt. geoSTATIONARY = 1-day + over equator
higher = slower (bigger circle, weaker gravity)

job table:

type	job (5 words)	example
navigation	tells you where you are	GPS
comms	bounces signals across Earth	Starlink
weather	watches clouds from GEO	GOES
…	…	…

🧠 SUMMARY — video closed, from memory (5 lines)

1. _____
2. _____
3. _____
4. _____
5. _____

FEEDBACK · YOUR PAGE

What was right, what to sharpen

Honest read of the photo you sent. The structure was correct on the first try, which is the hard part.

✔ Correct

- **Cues are questions.** "What are the types of satellites?" not "Satellite types." That's the thing most people never do.
- **Keywords, not sentences.** Nothing to fix there.
- **All three altitude ranges are right.** LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km.
- **You started a third cue mid-lecture** ("job of each satellite"). That's exactly how the column is supposed to grow.

▲ Sharpen

- **Geosynchronous ≠ geostationary.** Geosynchronous = the orbit takes one day, at any tilt. Geostationary = one day AND over the equator, so it hangs over one spot. TV and weather sats are geostationary. Write both words. Classic quiz trap.
- **Add the one "why."** GEO is 35,786 km because at that height one orbit takes exactly one Earth rotation (23 h 56 min). Lower = faster, higher = slower. Every other orbit number you'll just memorize; this one you can explain.
- **Cues are too easy.** "What are the orbit types?" you can answer with half a glance. Use cues you might fail: "Where does GPS live?", "Why 35,786?", "Higher = faster or slower?"
- **Eight-item list → four groups.** Flat lists of 8 don't retrieve. "Look out / look down / relay / live" does.
- **Job table = type · job · one real example.** The example is what sticks.

🖍 Left page was blank

This lecture is a picture: Earth, three rings, residents on each ring. Page 1 shows it. Draw yours badly first, then fix it. Don't copy page 1 neatly — the drawing is the memory, the copying isn't.

🧠 Missing: header + summary

- Top of page: section / lecture / date, so the index can point here.
- Bottom: five lines from memory, video closed. Can't fill five? That's your first gap. Good — now you know where it is.

📒 For the Question bank (back pages)

- Why is GEO exactly 35,786 km?
- Geosynchronous vs geostationary: which one hangs over one spot?
- Name the orbit and altitude of: ISS, GPS, a TV satellite.
- Group the 8 satellite types by job.

⚡ Next step: finish the lecture, write the 5-line summary, send me a photo. I'll check it and write the next cards.